

Experiment 1: Probability Theory

Aim

To calculate the prior probability of Malignant and Benign classes using the Breast Cancer Wisconsin Diagnostic Dataset and predict the class of a new data instance using Gaussian Naive Bayes.

Program

```
from sklearn.datasets import load_breast_cancer
from sklearn.model_selection import train_test_split
from sklearn.naive_bayes import GaussianNB

data = load_breast_cancer()

X = data.data
y = data.target

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

malignant_prob = sum(y_train == 0) / len(y_train)
benign_prob = sum(y_train == 1) / len(y_train)
print("Prior Probability of Malignant:", malignant_prob)
print("Prior Probability of Benign:", benign_prob)

model = GaussianNB()
model.fit(X_train, y_train)

new_instance = X_test[0].reshape(1, -1)
prediction = model.predict(new_instance)
probability = model.predict_proba(new_instance)

print("\nPosterior Probabilities")
print("Malignant:", probability[0][0])
print("Benign:", probability[0][1])
```

```
if prediction[0] == 0:  
    print("Predicted Class: Malignant")  
else:  
    print("Predicted Class: Benign")
```

Output



```
PS C:\personal\computer vision> & C:\Users\suman\AppData\Local\Microsoft\WindowsApps\python3.12.exe "c:/pe  
rsonal/computer vision/col-at3 1"  
Prior Probability of Malignant: 0.37142857142857144  
Prior Probability of Benign: 0.6285714285714286  
  
Posterior Probabilities  
Malignant: 2.42625722558841e-09  
Benign: 0.9999999975737435  
Predicted Class: Benign  
PS C:\personal\computer vision>
```

Result

Bayes Decision Theory successfully classified breast cancer cases by calculating prior and posterior probabilities, demonstrating effective probabilistic classification.